Implementation of the smooth activation of the source term in the enthalpy and species composition transport equations

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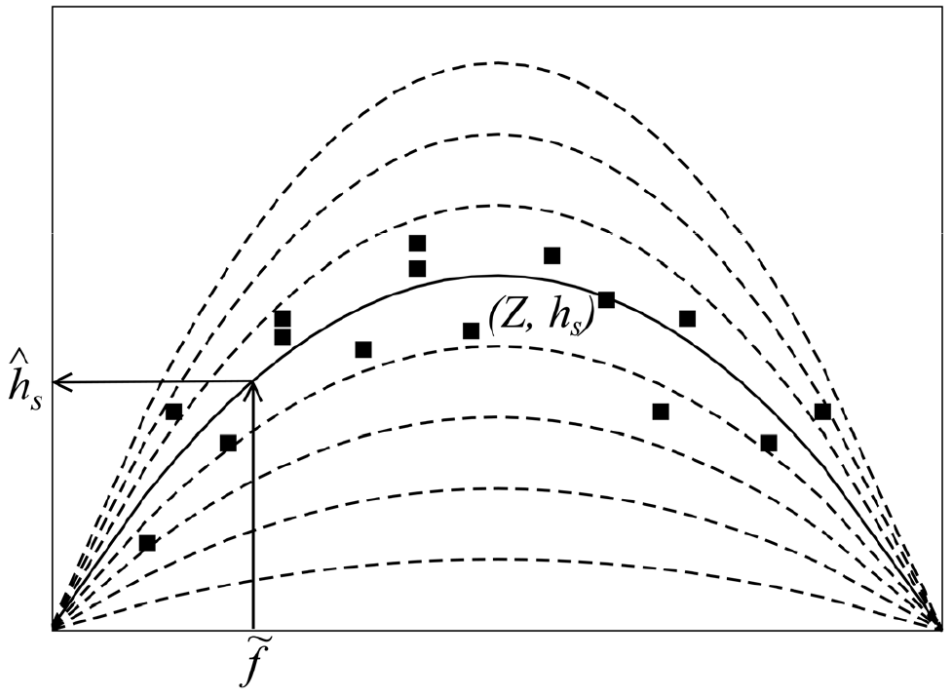
Background

Coupling between Eulerian LES scheme and Lagrangian MC particle scheme is archived by a conditional form of the equivalent enthalpy method suggested by Muradoglou et al. (2001) and enhanced by Raman et al. (2005):

- **Pre-computed flamelet equation solutions with different scalar dissipation rates**
- **Minimization of variance of the equivalent enthalpy between instantaneous particle data and pre-computed flamelet curves within a supercell**
- **Set equivalent enthalpy at LES cell in dependence on the LES mixture fraction value**

$$\frac{\partial \overline{\rho}\widetilde{h}_s^E}{\partial t} + \frac{\partial \overline{\rho}\widetilde{u}_i\widetilde{h}_s^E}{\partial x_i} = -\frac{\partial}{\partial x_i}\left(\widetilde{J}_{h_{s,i}} + J^{\mathrm{sf}}_{h_{s,i}}\right) + \boxed{\overline{\rho}\frac{\hat{h}_s - \widetilde{h}_s^E}{\tau_{\mathrm{rel}}}}$$



Source: M.J. Cleary, A.Y. Klimenko (2009) "A detailed quantitative analysis of sparse-Lagrangian filtered density function simulations in constant and variable density reacting jet flows", Physics of Fluids 23, 115102.

Background

Source term in equivalent enthalpy and species composition (not shown here) transport equation is relaxed by the relaxation time scale:

$$\frac{\partial \overline{\rho}\widetilde{h}_s^E}{\partial t} + \frac{\partial \overline{\rho}\widetilde{u}_i\widetilde{h}_s^E}{\partial x_i} = -\frac{\partial}{\partial x_i}\left(\widetilde{J}_{h_{s,i}} + J^{\mathrm{sf}}_{h_{s,i}}\right) + \boxed{\overline{\rho}\frac{\hat{h}_s - \widetilde{h}_s^E}{\tau_{\mathrm{rel}}}}$$

Thus, the relaxation time scale determines how strong the Eulerian LES field is correlated to the Lagrangian particle field. Following Cleary and Klimenko (2011) excessively small values can cause instabilities, while the two schemes would get de-correlated by using excessively large values for the relaxation time scale.

By default the relaxation time step should be set to approximately ten times the characteristic numerical time step.

Purpose

In some cases, foremost in turbulent reactive jet simulations, one cannot start the simulation from scratch with the full set of target curves, due to excessive occurrence of puffing effects in the flow field, which then lead to strong pressure waves propagating throughout the computational domain.

Thus, to avoid these waves it is recommended to take only one target curve for the early part of initial transient simulation time and to later switch to the full set of target curves.

But switching from one target curve to the full set instantly changes the value of the source term resulting in one big pressure wave.

The solution for this problem is a smooth activation of the source time wire the smooth reduction of the relaxation time scale from a large start value towards a target value:

$$\tau_{rel} = \max\left(\tau_{rel}^{start} - n\,\tau_{rel}^{delta}, \tau_{rel}^{target}\right).$$

For a negligible heat release, it might be the case, that a density coupling is not needed at all. Therefore additionally a switch is introduced, which completely deactivates the density coupling.

OF Modifications

src/lagrangian/mmc/clouds/Templates/ThermoPopeCloud/ ThermoPopeCloud.H

class ThermoPopeCloud definition

Necessary variables are introduced.

```
        Switch radiation_;

        //- Thermophysical coupling
+       Switch densityCouplingEnabled_;
        boolList solveEqvSpecie_;
        word primarySpecies_;
        Switch independentYEqvCurve_;
        List<label> TEqvTargetCurve_;
        List<label> YEqvTargetCurve_;
-       dimensionedScalar tauRelax_;
+       Switch tauRelaxBlending_;
+       dimensionedScalar tauRelaxStart_;
+       dimensionedScalar tauRelaxTarget_;
+       scalar tauRelaxDelta_;
        word tauUnits_;

    // Protected Member Functions
```

OF Modifications

src/lagrangian/mmc/clouds/Templates/ThermoPopeCloud/ ThermoPopeCloud.H

class ThermoPopeCloud definition

```
        inline const volScalarField& z() const;

        //- Return const access to thermophysical coupling data
+       inline const Switch& densityCouplingEnabled() const;
        inline const boolList& solveEqvSpecie() const;
        inline const List<label>& TEqvTargetCurve() const;
        inline const List<label>& YEqvTargetCurve() const;
-       inline const dimensionedScalar& tauRelax() const;
+       inline const dimensionedScalar& tauRelaxStart() const;
+       inline const dimensionedScalar& tauRelaxTarget() const;
+       inline const scalar& tauRelaxDelta() const;
        inline const word& tauUnits() const;
```

Necessary accessor functions are defined.

OF Modifications

src/lagrangian/mmc/clouds/Templates/ThermoPopeCloud/ ThermoPopeCloud.C

In function *setThermoPhysicalCoupling*

```
     const dictionary couplingDict(this->cloudProperties().subDict("thermophysicalCoupling"));
+
+    densityCouplingEnabled_ = readBool(couplingDict.lookup("enabled"));
+
+    Info << "Density coupling: " << densityCouplingEnabled_ << endl;

     forAll(composition().componentNames(), ns)
     {
```

Logical switch is read in and printed in the standard output.

OF Modifications

src/lagrangian/mmc/clouds/Templates/ThermoPopeCloud/ ThermoPopeCloud.C

In function *setThermoPhysicalCoupling*

```
        independentYEqvCurve_ = false;
    else
        independentYEqvCurve_ = true;
+
+   tauRelaxBlending_ = couplingDict.lookupOrDefault<Switch>("tauRelaxBlending",false);
```

Logical switch for the blending.

```
-   const scalar tauTmp(readScalar(couplingDict.lookup("tauRelax")));
+   const scalar tauTmpTarget(readScalar(couplingDict.lookup("tauRelax")));
```

```
-   tauRelax_.value() = tauTmp;
+   tauRelaxTarget_.value() = tauTmpTarget;
+
+   if (tauRelaxBlending_)
+   {
+       const scalar tauTmpStart(readScalar(couplingDict.lookup("tauRelaxStart")));
+
+       tauRelaxStart_.value() = tauTmpStart;
+
+       const scalar tauTmpDelta(readScalar(couplingDict.lookup("tauRelaxDelta")));
+
+       tauRelaxDelta_ = tauTmpDelta;
+   }
+   else
+   {
+       tauRelaxStart_.value() = tauTmpTarget;
+       tauRelaxDelta_ = 1;
+   }
```

A start and a target value are read in. The blending step is given by a given delta.

Default values ensure backward compatibility.

```
    const word tUnits(couplingDict.lookup("tauUnits"));
```

OF Modifications

src/lagrangian/mmc/clouds/Templates/ThermoPopeCloud/ ThermoPopeCloud.C

In the instantiation after the *ThermoPopeCloud* constructor

```
     TEqvTargetCurve_(),

     YEqvTargetCurve_(),
+
+    tauRelaxStart_("tauRelaxStart",dimTime,1),

-    tauRelax_("tauRelax",dimTime,1),
+    tauRelaxTarget_("tauRelaxTarget",dimTime,1),
+
+    tauRelaxDelta_(1),

     tauUnits_()
```

Initialization of variables at constructor call.

OF Modifications

src/lagrangian/mmc/clouds/Templates/ThermoPopeCloud/ ThermoPopeCloud.C

In function *EqvETargetValues*

```
template<class CloudType>
void Foam::ThermoPopeCloud<CloudType>::EqvETargetValues
(
    const volScalarField& f,
    PtrList<volScalarField>& YEqvETarget,
    volScalarField& TEqvETarget
)
{
+    if (!densityCouplingEnabled())
+        return;
+
     //- Find target curves
     findEquivalentTempSpeciesCurves();
```

Exit function immediately if density coupling is switched off.

OF Modifications

src /applications/solvers/mmc/mmcFoam/hYEqvE_Eqn.H

```
-pSets.first()->EqvETargetValues(f,YEqvETarget,TEqvETarget);
+if (pimple.firstIter())
+{
+    pSets.first()->EqvETargetValues(f,YEqvETarget,TEqvETarget);
+}
```

```
hEqvETarget = thermoEqvETarget.he(p,TEqvETarget);
```

Since particle information does not change between two pimple loops in *mmcFoam*, the determination of the target curves is only done once that happens in the first one.

OF Modifications

src /applications/solvers/mmc/mmcFoam/hYEqvE_Eqn.H

```
 dimensionedScalar tauRel("tauRel", dimTime, 0);

-if (pSets.first()->tauUnits() == "time")
+label deltaIndices = mesh.time().timeIndex() - mesh.time().startTimeIndex() - 1;
+
+tauRel.value() = max(pSets.first()->tauRelaxStart().value() - deltaIndices*pSets.first()->tauRelaxDelta(), pSets.first()->tauRelaxTarget().value());
+
+if (pSets.first()->tauUnits() == "timestep")
 {
-    tauRel.value() = pSets.first()->tauRelax().value();
+    tauRel.value() = tauRel.value()*mesh.time().deltaT().value();
 }
```

Here the actual blending is done via formula:

$$\tau_{rel} = \max\left(\tau_{rel}^{start} - n\,\tau_{rel}^{delta}, \tau_{rel}^{target}\right).$$

Here n is the number of time steps.

OF Modifications

src /applications/solvers/mmc/mmcFoam/hYEqvE_Eqn.H

```
+if (pSets.first()->tauRelaxBlending())
+{
+    Info << "tauRealaxStart = " << pSets.first()->tauRelaxStart().value() << endl;
+    Info << "tauRelaxTarget = " << pSets.first()->tauRelaxTarget().value() << endl;
+    Info << "tauRelaxDelta = " << pSets.first()->tauRelaxDelta() << endl;
+    Info << "tauRealax = " << tauRel.value() << endl;
+}
+
+
 if (tauRel.value() < mesh.time().deltaT().value())
     tauRel.value() = mesh.time().deltaT().value();
```

In case of an active blending some information are given in the standard output.

OF Modifications

src /applications/solvers/mmc/mmcFoam/hYEqvE_Eqn.H

```
forAll(YEqvE, specieI)
{
    if (YEqvE[specieI].name() != inertSpecie)
    {
        volScalarField& Yi = YEqvE[specieI];

        volScalarField& YTarget = YEqvETarget[specieI];
+
+       if (!pSets.first()->densityCouplingEnabled())
+           YTarget = Yi;

        if (!pSets.first()->solveEqvSpecie()[specieI])
        {
```

If density coupling is swiched off, the difference within the source term is set to zero which sets the whole source term to zero.

How to use: Tutorials

Blending is set in the *thermophysicalCoupling dictionary* in file constant/cloudProperties:

```
 //- ThermophysicalModel coupling
 thermophysicalCoupling
 {
+       enabled                 true;
+
     //- List of species for which equivalent
     //  mass fraction solutions are obtained
     //  on the FV grid
     CH4;
     O2;
     CO2;
     H2O;

     primarySpecies  CO2;
-
-    tauRelax 10;
-    tauUnits timestep;
+
+       tauRelaxBlending true;
+
+       tauRelax                10;
+       tauRelaxStart   14;
+       tauRelaxDelta   2;
+
+    tauUnits            timestep;
 }
```

No other input data needed.